ALGEBRA 1 · RELATIONS AND FUNCTIONS

Function Detector

Explore & Reflect TEACHER KEY

Name _____ Date _____ Period _____

Open **Function Detector** on Mr. K's Classroom: Algebra 1 → Relations and Functions → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Do you think a circle is a function? Why or why not?

   No. A vertical line through a circle hits it in two points, so one input has two outputs.

2. A relation contains (1, 4) and (1, 7). Is it a function?

   No. The input 1 has two different outputs.

2 Explore use the tool and record what you see

Sweep each relation

Drag the vertical line all the way across. Domain and range appear after a full sweep.

Relation	Passes the vertical line test?	Function?	Domain	Range
Line	**yes**	**yes**	**all real numbers**	**all real numbers**
Parabola	**yes**	**yes**	**all real numbers**	**$y \geq -4$**
Sideways parabola	**no**	**no**	**$x \geq -3$**	**all real numbers**
Circle	**no**	**no**	**$-5 \leq x \leq 5$**	**$-5 \leq y \leq 5$**
Absolute value	**yes**	**yes**	**all real numbers**	**$y \geq -3$**
Vertical line	**no**	**no**	**$x = 3$ only**	**all real numbers**

My points

Plot points by clicking the grid. Watch the mapping diagram.

	Your points	What the mapping diagram shows
A function	**answers vary; no x-value repeats**	**each input has one arrow**
Not a function	**answers vary; some x-value repeats with a different y**	**one input has two arrows (red)**

3 Write the rule fill in each blank from what you observed

1. A relation is a function when each **input (x)** has exactly one **output (y)**.
2. If any vertical line hits a graph in more than one point, the graph **is not** a function.
3. In a mapping diagram of a function, no input has **two arrows (two outputs)**.
4. The domain is the set of all **inputs (x-values)**. The range is the set of all **outputs (y-values)**.

4 Check no tool: show your work

1. Is {(2, 3), (4, 3), (6, 1)} a function?

   Yes. Outputs may repeat; each input has one output.

2. Is {(–1, 0), (2, 5), (–1, 4)} a function?

   No. –1 has two outputs.

3. Give the domain and range of $y = x^2 + 1$.

   Domain: all real numbers. Range: $y \geq 1$.

4. Is $x = y^2$ a function?

   No. For example, $x = 4$ gives $y = 2$ and $y = -2$.

5 Reflect

Why does the vertical line test work?

A vertical line marks one x-value. If it hits the graph twice, that x-value has two y-values, so it isn't a function.